
Averaged Jacobian Regularity Can Misrank Few-Step Flow-Matching Schedules: A Certified Gaussian Counterexample

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Abstract

1 Few-step sampling in flow matching and stochastic interpolants requires an inter-
2 polation schedule before any endpoint error is observed. Averaged squared
3 Jacobian regularity is a plausible schedule-design criterion, but it is not a solver-
4 specific discretization-error functional. Chen et al. propose minimizing A_2 , the
5 time-integrated squared spatial Jacobian Lipschitzness of a flow-matching marginal
6 ODE. They do not claim or prove that A_2 universally ranks equal-NFE solver error.
7 For independent $N(0, 1)$ and $N(0, 4)$, exact regularity integrals prefer trigonomet-
8 ric VP while explicit Heun, a two-stage Runge–Kutta method, at eight number of
9 function evaluations (NFE 8) prefers the linear path in Gaussian W_2 . The inversion
10 is certified by an exact rational Heun product and a nonnegative polynomial in π
11 and $\sqrt{2}$. A grid-aware construction shows that, for every step count N , the unique
12 integrated-regularity minimizer can have strictly larger N -step Euler error than a
13 higher-regularity competitor aligned to that grid. A specified finite Gaussian enu-
14 meration of four candidate paths on 36 tested blocks contains both agreement and
15 pairwise disagreement. The results establish a limitation of a universal surrogate
16 interpretation. They do not evaluate learned fields and do not estimate a population
17 frequency of failure.

18 1 Introduction

19 Few-step sampling commits to a path, a solver, and a small NFE budget before any error is observed.
20 Lipschitz constants enter classical one-step bounds [Hairer et al., 1993]. Chen et al. [2025] propose
21 minimizing A_2 , the time-integrated squared spatial Jacobian norm (their Definition 3.2), as a schedule-
22 design objective in a specified admissible family. They do not claim or prove that A_2 universally
23 ranks equal-NFE solver error. We study that narrower surrogate question on a flow-matching marginal
24 ODE.

25 The object is an integrated unsigned Jacobian energy. For a fixed finite-step solver, endpoint error
26 depends on stagewise field values, temporal derivatives, quadrature structure, and signed cancellations
27 that this energy does not determine. For independent $N(0, 1)$ and $N(0, 4)$, exact regularity prefers
28 trigonometric VP while Heun at NFE 8 prefers the linear path in Gaussian W_2 (Section 3). A
29 solver-grid-aware construction then shows that even the global $\mathcal{R}$ -minimizer need not minimize
30 fixed-grid Euler error (Section 6); the competitor is built from the solver resolution. A finite Gaussian
31 enumeration of four specified candidate paths on 36 tested blocks contains both agreement and
32 pairwise disagreement (Section 5). No learned field is used.

33 Venue relevance is through distribution learning, Wasserstein endpoint error, and the geometry
34 of Gaussian covariance interpolation, not through a new geometric architecture or a practical OT

algorithm. Tao and Choi [2026] give a broader dynamical error analysis, including material derivatives, strain, vorticity, logarithmic norms, and learned models. The present contribution is narrower but exact and auditable: a certified scalar Gaussian ranking inversion, a closed-form W_2 comparison, a grid-aware impossibility construction, and a transparent finite enumeration. Contemporaneous Lipschitz Euler rates [Stéphanovitch, 2026] bound a different object.

2 Gaussian interpolants and averaged regularity

Let $X_0 \sim N(\mu_0, \Sigma_0)$ and $X_1 \sim N(\mu_1, \Sigma_1)$ be independent. A shared schedule is a scalar pair (α, σ) . We distinguish three regularity classes.

- Class S: $\alpha, \sigma \in C^1([0, 1])$ with $\alpha(0) = \sigma(1) = 1$, $\alpha(1) = \sigma(0) = 0$, and $(\alpha(t), \sigma(t)) \neq (0, 0)$ on $[0, 1]$. Linear and trigonometric VP belong to Class S.
- Class Q: a C^1 path of SPD covariances $Q(t)$ with continuous $A = \frac{1}{2}Q'Q^{-1}$. The per-mode schedule $q_i(t) = \lambda_i^t$ is Class Q.
- Class R: $\alpha^2, \sigma^2 \in C^1([0, 1])$, $\alpha^2 \geq 0$, $\sigma^2 \geq 0$, $\alpha^2 + \sigma^2 > 0$, and $\alpha(0)^2 = 1$, $\sigma(0)^2 = 0$, $\alpha(1)^2 = 0$, $\sigma(1)^2 = 1$. Square roots need not be C^1 at the endpoints. Chen Ex. 3.3, $\sigma(t)^2 = (M^t - 1)/(M - 1)$ and $\alpha(t)^2 = (M - M^t)/(M - 1)$ for $M > 0$, $M \neq 1$, is Class R and Class Q, not Class S. The $M = 1$ limit is $\sigma(t)^2 = t$, $\alpha(t)^2 = 1 - t$.

The interpolant $X_t = \alpha X_0 + \sigma X_1$ [Albergo et al., 2025, Lipman et al., 2023] is Gaussian with mean $m_t = \alpha\mu_0 + \sigma\mu_1$ and covariance $Q = \alpha^2\Sigma_0 + \sigma^2\Sigma_1$. The marginal field is affine,

$$b(t, x) = A(t)x + c(t), \quad A = C_t Q^{-1}, \quad c = \dot{m} - A m. \quad (1)$$

The formula $C_t = \frac{1}{2}(\alpha^2)' \Sigma_0 + \frac{1}{2}(\sigma^2)' \Sigma_1$ extends continuously to $[0, 1]$ for Class R; equivalently $C_t = \dot{\alpha}\alpha \Sigma_0 + \dot{\sigma}\sigma \Sigma_1$ on $(0, 1)$ where the coefficients are differentiable and nonzero. This is a flow-matching marginal ODE, not a score-based probability-flow ODE in the sense of Song et al. [2021]. If $\Sigma_0 = I$, then Q , Q' , A , and Σ_1 share the eigenbasis of Σ_1 , and mode i obeys $x'_i = a_i(t)x_i + c_i(t)$ with $q_i = \alpha^2 + \lambda_i\sigma^2$ and $a_i = q'_i/(2q_i)$. The covariance drift formula is independent of the mean path $c(t)$. We compare the linear path $(1 - t, t)$ and trigonometric VP $(\cos(\pi t/2), \sin(\pi t/2))$. Following Chen et al. [2025, Definition 3.2], $\mathcal{R}[b] = \int_0^1 \|A(t)\|_2^2 dt$. For state-independent A this coincides with their path expectation. By definition $\mathcal{R}$ ignores $c(t)$. Error is Gaussian W_2 of the endpoint laws [Gelbrich, 1990]. This is neither pathwise trajectory error nor local truncation error. Equal NFE uses $S = B/s$ steps with $s = 1, 2, 4$ for Euler, Heun, and RK4.

Any regular variance path from 1 to $\lambda > 0$ obeys $\int_0^1 a dt = \frac{1}{2} \ln \lambda$. Hence $\mathcal{R} = \int_0^1 a^2 dt = (\int a)^2 + \text{Var}_t(a) \geq (\frac{1}{2} \ln \lambda)^2$, with equality if and only if a is constant. The one-dimensional minimizer is $q(t) = \lambda^t$, $a \equiv \frac{1}{2} \ln \lambda$, which is Chen et al. Example 3.3. Linear and trigonometric VP are C^1 shared interpolants. In several dimensions, the per-mode schedule $q_i(t) = \lambda_i^t$ yields $\mathcal{R} = (\frac{1}{2} \max_i |\ln \lambda_i|)^2$. When Σ_1 has at least two distinct eigenvalues, that path is not a Class S shared interpolant.

3 A one-dimensional Gaussian ranking counterexample

Assumption 1. Endpoints $N(0, 1)$ and $N(0, 4)$ are independent. Schedules are linear and trigonometric VP. The solver is explicit Heun with NFE 8 (four steps of size 1/4).

Proposition 1. Under the assumption, $\mathcal{R}_{\text{lin}} = 5\pi/8 - 1 > \mathcal{R}_{\text{VP}} = \pi^2/16$, while the Heun factors satisfy $r_{\text{lin}} = 6797469/3559400$ and $0 < r_{\text{VP}} < 187/100$. Hence $W_2^{\text{lin}} < 0.091 < 0.130 < W_2^{\text{VP}}$.

Proof sketch. The linear variance is $q_{\text{lin}}(t) = 1 - 2t + 5t^2$. Completing the square gives the equivalent identities $q_{\text{lin}}(t) = 5[(t - 1/5)^2 + (2/5)^2] = 5(t - 1/5)^2 + 4/5$. The substitution $u = t - 1/5 = (2/5) \tan \theta$ converts $\int a_{\text{lin}}^2$ into $(5/2) \int \sin^2 \theta d\theta$. Limits $\arctan(-1/2)$ and $\arctan 2$ together with $\arctan 2 + \arctan(1/2) = \pi/2$ yield $\mathcal{R}_{\text{lin}} = 5\pi/8 - 1$. The VP integral evaluates to $\pi^2/16$ by a Weierstrass identity. The comparison $5\pi/8 - 1 > \pi^2/16$ rearranges to $(\pi - 2)(\pi - 8) < 0$, which holds because $2 < \pi < 8$. The four linear Heun factors are the rationals $47/52$, $6/5$, $497/370$, and $97/74$, all positive. Exact VP grid values are $a(0) = a(1) = 0$, $a(1/2) = 3\pi/10$, and $a(1/4) =$

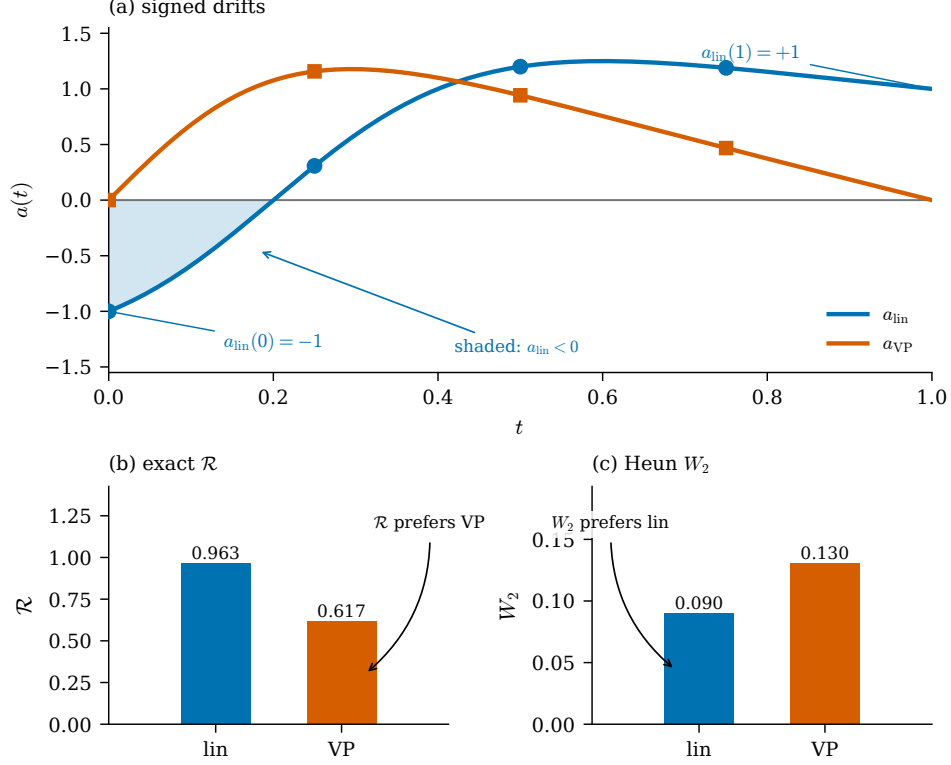


Figure 1: Certified scalar counterexample for independent $N(0, 1)$ and $N(0, 4)$: drifts, exact $\mathcal{R}$ integrals, and Heun NFE 8 Gaussian W_2 .

81 $(9 + 15\sqrt{2})\pi/82$, $a(3/4) = (-9 + 15\sqrt{2})\pi/82$. Positivity of $a(3/4)$ is $15^2 \cdot 2 - 9^2 = 369 > 0$. The
82 four-step product lies in $\mathbb{Q}[\pi, \sqrt{2}]$ with nonnegative coefficients, so substituting $\sqrt{2} < 99/70$ and
83 $\pi < 355/113$ is valid and yields $r_{\text{VP}} < 187/100$ by an integer comparison in Appendix A. Interval
84 arithmetic is a cross-check only. $\square$

85 4 Sign change and transported defects

86 For any independent linear coupling, $a_{\text{lin}}(0) = -1 < 0 < a_{\text{lin}}(1) = +1$. The VP drift vanishes at
87 the endpoints and does not change sign on $(0, 1)$. Averaging a^2 does not retain that sign change.
88 Fixed-stage Runge–Kutta methods use signed stage evaluations that are not determined by $\mathcal{R}$. The
89 sign difference is consistent with the observed inversion, but the analysis does not isolate it as a causal
90 contribution.

91 Let $e_n = \sqrt{q(t_{n+1})/q(t_n)}$ and let r_n be the numerical factor. The telescoping identity

$$\prod_n r_n - \prod_n e_n = \sum_n (r_n - e_n) \left(\prod_{j < n} r_j \right) \left(\prod_{j > n} e_j \right) \quad (2)$$

92 is an accounting tool, not the content of the inversion. The content is the leading exact-minus-method
93 coefficient: $\frac{1}{2}(a' + a^2)h^2$ for Euler and $(-a''/12 + a^3/6)h^3$ for the implemented Heun factor. Those
94 expansions involve different derivatives, signs, and stage locations.

95 5 Four-path enumeration and the inversion region

96 Source $N(0, I)$. Four centered targets: anisotropic Gaussians with condition number 4 in dimensions
97 2 and 8 (eigenvalues a geometric sequence from $1/2$ to 2); factor-plus-noise covariances $\Sigma =$
98 $FF^\top + 0.05I$ with $F \in \mathbb{R}^{d \times 2}$ i.i.d. standard normal (seed 271828). When $d = 2$ the factor rank

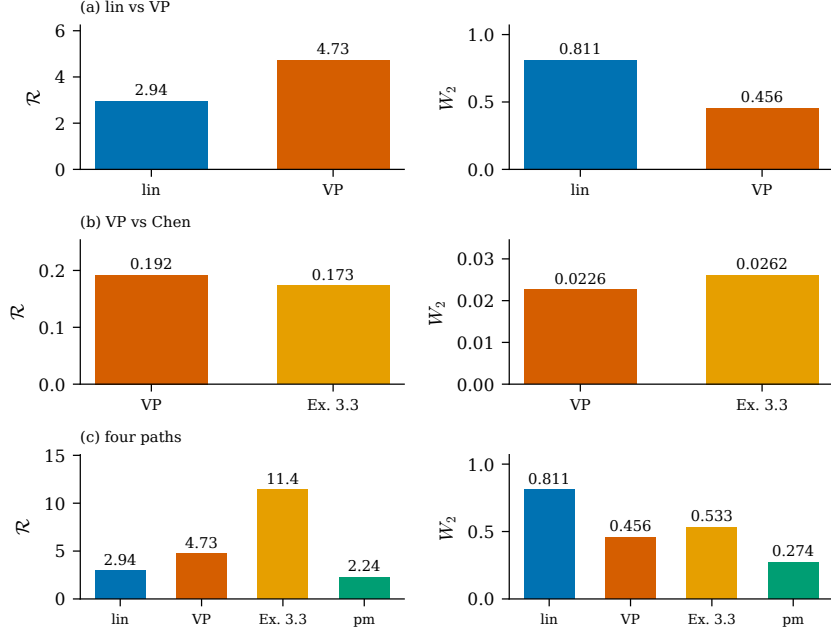


Figure 2: Three comparisons. Left: $\mathcal{R}$. Right: Euler NFE 8 W_2 . (a) linear vs VP, rank-2 factor-plus-noise $d = 8$; (b) VP vs Chen Ex. 3.3, anisotropic $d = 8$; (c) four specified paths, rank-2 factor-plus-noise $d = 8$. Among those four paths, per-mode wins $\mathcal{R}$ and W_2 in 36 of 36 tested blocks.

Table 1: Four specified schedules at Heun, NFE 8. Among these four candidate paths, per-mode has the smallest W_2 on every geometry in the 36 tested blocks. When Σ_1 has at least two distinct eigenvalues, the per-mode path is not a C^1 shared interpolant.

family	d	$\mathcal{R}_{\text{lin}}$	$\mathcal{R}_{\text{VP}}$	$\mathcal{R}_{\text{sc}}$	$\mathcal{R}_{\text{log}}$	Heun-8 W_2 min
anisotropic	2	0.972	0.192	0.173	0.120	log-cov
anisotropic	8	0.972	0.192	0.173	0.120	log-cov
factor+noise	2	1.246	0.625	0.668	0.380	log-cov
rank-2 factor+noise	8	2.944	4.731	11.419	2.244	log-cov

99 equals the ambient dimension. Because $\mathcal{R}$ uses the spectral norm, the two anisotropic dimensions
 100 share the same $\mathcal{R}$ pair (0.972296, 0.192270), so the grid contains three distinct linear-versus-VP $\mathcal{R}$
 101 comparisons. Each geometry is run under Euler, Heun, and RK4 at NFE $\{8, 16, 32\}$. Headline $\mathcal{R}$ is
 102 the continuous integral of $\max_i a_i(t)^2$. Ranking inversions were identified in Phase 3; the counts
 103 below are a confirmatory Phase 4 reproduction on configurations frozen before the rerun.

104 The per-mode log-covariance path $q_i(t) = \lambda_i^t$ is one of four specified candidate paths. Among those
 105 four paths on the stated 36 tested blocks it has the smallest $\mathcal{R}$ on every geometry and the smallest
 106 Gaussian W_2 in all 36 tested blocks (Table 1 reports Heun at NFE 8). This finite enumeration
 107 does not imply that a global $\mathcal{R}$ -minimizer minimizes fixed-NFE error. When Σ_1 has at least two
 108 distinct eigenvalues, that path is not a C^1 shared interpolant. Linear versus VP invert in 5 of 12
 109 geometry $\times$ solver cells; 4 of those 12 invert at every tested NFE. As a post-hoc pairwise comparison
 110 within the shared-schedule class, trigonometric VP versus Chen Ex. 3.3 with $M = \lambda_{\max}$ invert in 9
 111 of 36 tested blocks and 4 of 12 cells.

112 A one-dimensional sweep $\lambda \in [0.05, 100]$ (Figure 3) locates the linear-versus-VP inversions. At $\lambda =$
 113 4, the inversion is present under Heun at every tested budget NFE $\in \{4, 8, 12, 16, 24, 32, 64, 128, 256\}$,
 114 absent under Euler, and mixed under RK4 (absent at NFE 8 and 12). Euler *does* invert at more
 115 extreme λ on the same sweep. The scalar example is therefore solver-dependent, not an artefact of
 116 Heun at budget 8 alone.

Table 2: Strongest centered inversion: rank-2 factor-plus-noise, $d = 8$, Euler, NFE 8. Eigenvalues: 0.05 (six times), 4.64827725629505, 10.675638226546441. Deterministic calculation: $\mathcal{R}$ is evaluated by adaptive quadrature, while W_2 uses exact Gaussian moments propagated through the modal Runge–Kutta updates.

path	$\mathcal{R}$	$\hat{\mathcal{R}}_{24}$	Gaussian W_2
linear	2.9441044083	2.9476523251	0.8108540111
VP	4.7305438136	4.7295206355	0.4564779075

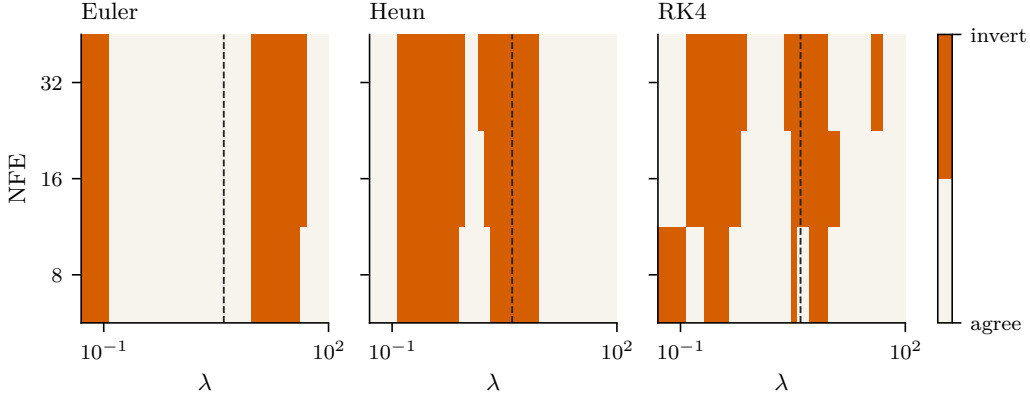


Figure 3: Linear versus VP ranking inversion versus target variance λ . Dark cells: $\mathcal{R}$ and W_2 disagree. The dashed line is $\lambda = 4$ (Proposition 1).

6 A grid-aware Class S Euler construction

Theorem 1. Fix $L > 0$ and an integer $N \geq 1$. Let $h = 1/N$ and let forward Euler use left endpoints $t_n = nh$ on $[0, 1]$. Set $\varepsilon_N = N(e^{L/N} - 1) - L > 0$, $a_0(t) \equiv L$, and $a_{1,N}(t) = L + \varepsilon_N \cos(2\pi Nt)$. For $x' = a(t)x$, $x(0) = 1$, and the Class S embeddings $q_k = \exp(2 \int_0^t a_k)$, $\alpha_k = \sqrt{q_k} \cos(\pi t/2)$, $\sigma_k = \sqrt{q_k/e^{2L}} \sin(\pi t/2)$: both exact endpoints equal e^L ; a_0 uniquely minimizes $\int a^2$ among scalar fields with integral L ; Euler is exact for $a_{1,N}$ and strictly inexact for a_0 . Endpoint error is $|r - e^L|$, equivalently Gaussian W_2 between $N(0, r^2)$ and $N(0, e^{2L})$.

The competitor is constructed as a function of the solver grid. Phase offsets and non-grid frequencies destroy exact Euler aliasing. This is a solver-grid-dependent impossibility result, not a model of natural schedules, and not the mechanism of Proposition 1.

7 Related work

Flow matching and interpolants supply the Gaussian family [Lipman et al., 2023, Albergo et al., 2025, Liu et al., 2023, Tong et al., 2024]. Lipschitz-guided design proposes the selection criterion tested [Chen et al., 2025]. Solver-specific schedules and exact linear parts are documented in diffusion practice [Sabour et al., 2024, Karras et al., 2022, Lu et al., 2022, Zhang and Chen, 2023]. Stéphanovitch [2026] prove sharp Lipschitz regularity of canonical flow-matching fields and Euler-type W_2 rates; that is a bound, not a pairwise ranking of two interpolants at equal NFE.

Tao and Choi [2026] decompose $\nabla v = S + \Omega$ and control Euler error by the logarithmic norm $\mu_2 = \lambda_{\max}(S)$, which is signed, whereas $\mathcal{R}$ integrates $\|A\|_2^2$. In one dimension $\mu_2(a) = a$ and $\|A\|_2 = |a|$. Their Theorem 5 states that McCann displacement interpolation has vanishing material derivative, hence Euler is order 2 in the Eulerian analysis; Remark 7 records the stronger Lagrangian exactness observed on that interpolant. Neither statement applies to the independent-coupling linear or VP path studied here. The present paper supplies a rationally certified Gaussian instance of ranking failure for the unsigned average $\|A\|_2^2$. A counterexample to μ_2 itself is outside our scope. Benton

et al. [2024] bound W_2 of a learned flow; that is an error bound, not a pairwise ranking. Consistency models [Song et al., 2023] distill few-step maps.

8 Limitations

The systems are commuting affine Gaussian interpolants, three fixed-step Runge–Kutta methods, and small equal-NFE budgets. No learned neural field is evaluated. Chen et al. do not claim or prove a discretization-error bound, nor that A_2 universally ranks equal-NFE solver error. Among four specified candidate paths, the per-mode path wins 36 of 36 tested blocks; that does not imply that a global $\mathcal{R}$ -minimizer minimizes fixed-NFE error. Theorem 1 is grid-aligned by construction. Cell counts are hierarchical descriptive summaries, not population rates.

9 Conclusion

An integrated unsigned Jacobian-energy functional does not contain enough solver-specific temporal information to universally determine few-step endpoint error. Averaged squared Jacobian regularity can misrank equal-NFE Gaussian W_2 already for two C^1 interpolants of independent $N(0, 1)$ and $N(0, 4)$. A grid-aware construction shows that even the global $\mathcal{R}$ -minimizer need not minimize fixed-grid Euler error. A specified finite enumeration contains both agreement (36 of 36 tested blocks among four candidate paths) and pairwise disagreement.

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198 A Rational VP Heun certificate

199 This appendix records the integer comparison used in Proposition 1. It is part of this PDF, not a
200 separate supplement file.

201 The four-step VP Heun product is the following element $P(\pi, \sqrt{2})$ of $\mathbb{Q}[\pi, \sqrt{2}]$, obtained by
202 expanding the four Heun factors with $h = 1/4$ and reducing $\sqrt{2}^2 = 2$:

$$P(\pi, \sqrt{2}) = 1 + \frac{3}{40}\pi + \frac{15}{164}\pi\sqrt{2} + \frac{78579}{10758400}\pi^2 + \frac{45}{5248}\pi^2\sqrt{2} + \frac{11421}{17213440}\pi^3 + \frac{4671}{17213440}\pi^3\sqrt{2} \\ + \frac{5103}{275415040}\pi^4 + \frac{567}{55083008}\pi^4\sqrt{2} + \frac{243}{2203320320}\pi^5 + \frac{729}{2203320320}\pi^5\sqrt{2} + \frac{729}{176265625600}\pi^6. \quad (3)$$

203 Each of the twelve coefficients is a ratio of nonnegative integers, hence nonnegative, and the constant
204 term is 1. Substituting $\pi < 355/113$ and $\sqrt{2} < 99/70$ is therefore valid and yields

$$P\left(\frac{355}{113}, \frac{99}{70}\right) = \frac{192113353671412470139303}{102753427879939708813312} < \frac{187}{100},$$

205 equivalently

$$19211335367141247013930300 < 19214891013548725548089344.$$

206 Hence $r_{\text{VP}} < 187/100$ and $W_2^{\text{VP}} > 13/100$. The bound $\sqrt{2} < 99/70$ is $99^2 - 2 \cdot 70^2 = 1$;
207 $\pi < 355/113$ follows from Machin’s formula with Leibniz remainders (seven Taylor terms for
208 $\arctan(1/5)$, four for $\arctan(1/239)$). An independent mpmath 1.3.0 interval enclosure is a
209 cross-check, not the proof.